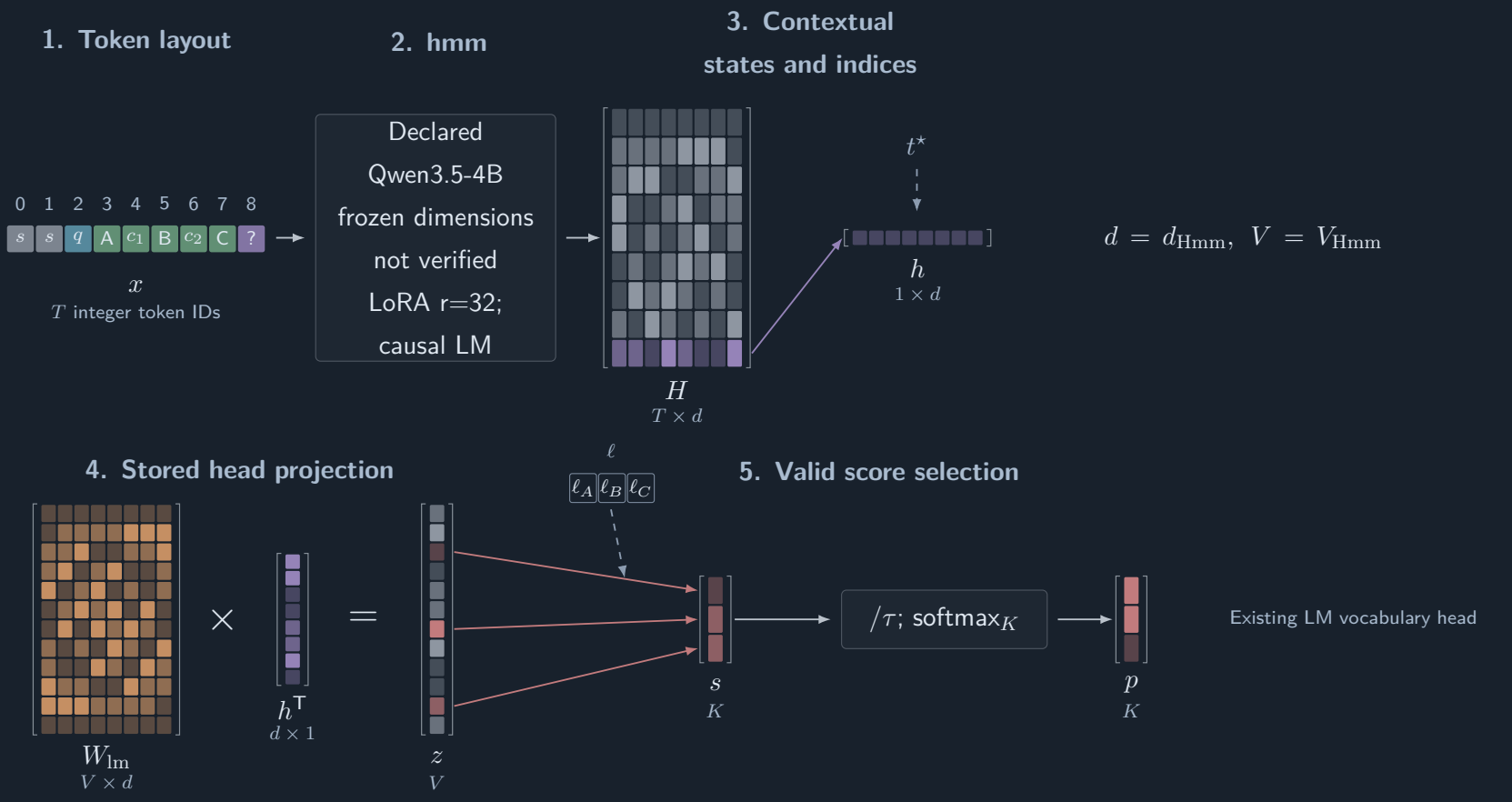
$$h = H_{t^\star,:}, \quad s_k = \langle W_{\ell_k,:}, h \rangle, \quad p = \mathrm{softmax}_K(s/\tau)$$



Axes	$d = d_{\mathrm{Hmm}}$, $V = V_{\mathrm{Hmm}}$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.
Objects	Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.
Mechanism	Shown: exact training-forward chain. Serving renormalizes a returned top-logprob subset; more than 26 options use chunks, not a single 255-slot head. The training path gathers A–Z from full vocabulary logits, masks invalid letters and mixes hard/soft CE. No new scalar head.